Macroeconomics I — MPE Insper

Practice list: General Equilibrium, Money and Inflation — graduate level (Listas 5–6)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — General equilibrium and the neoclassical growth model

(Kurlat ch. 9; Romer ch. 2 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A two-period economy has a fixed capital stock K and no investment. Households have $u(c) + \psi \ln l$ each period, $u = c^{1-\sigma}/(1-\sigma)$, one unit of time ($l + L = 1$); firms produce $Y = AK^{\alpha}L^{1-\alpha}$ in competitive markets. In equilibrium, with $L = 1/3$, $\sigma = 2$ and $\alpha = 1/3$, the elasticity $d\ln L/d\ln A$ is:

A. -0.46

B. 0

C. $+0.46$

D. -1.00

b) (3 points) In the continuous-time Ramsey model with $f(\tilde{k}) = \tilde{k}^{1/3}$, $\rho = 0.03$, $\sigma = 2$, $g = 0.02$, $n = 0.01$, $\delta = 0.05$, the steady-state gross saving rate $(n + g + \delta)\tilde{k}^*/f(\tilde{k}^*)$ is:

A. 33.3%

B. 22.2%

C. 26.7%

D. 16.7%

c) (2 points, T or F) In the Ramsey model, a permanent, unanticipated rise in government purchases financed by lump-sum taxes lowers consumption one for one on impact and leaves the capital stock unchanged forever.

d) (2 points, T or F) A permanent tax on capital income whose revenue is rebated lump-sum leaves the steady-state capital stock unchanged, because the rebate gives the revenue back.

Question 2. (15 points) Discursive. The Ramsey–Cass–Koopmans model. Households (population growing at n) maximize $\int_0^{\infty} e^{-\rho t}u(c)\,dt$, $u = c^{1-\sigma}/(1-\sigma)$, own the capital and rent it to firms; firms produce $Y = F(K, AL)$ with constant returns, A grows at g; capital depreciates at δ. Assume $\rho > n + (1-\sigma)g$. Notation: $\tilde{k} = K/(AL)$, $\hat{c} = c/A$, $f(\tilde{k}) = F(\tilde{k}, 1)$. Numbers: $f = \tilde{k}^{1/3}$, $\rho = 0.03$, $\sigma = 2$, $g = 0.02$, $n = 0.01$, $\delta = 0.05$.

a) (3 points) Derive the Euler equation from the discrete-time condition $u'(c_t) = e^{-\rho\Delta}(1 + r_{t+\Delta}\Delta)u'(c_{t+\Delta})$ (periods of length Δ) by letting $\Delta \to 0$. Use the firms' problem and no-arbitrage to get $r = f'(\tilde{k}) - \delta$, and write the system in $(\tilde{k}, \hat{c})$.

Solution:

b) (4 points) Prove that the interior steady state is unique and that $\tilde{k}^*$ lies below the Golden Rule. Compute $\tilde{k}^*$, $\hat{c}^*$, r^*, the saving rate and $\tilde{k}_{gold}$.

Solution:

c) (4 points) Linearize the system around the steady state. Show that it is a saddle, give the eigenvalues, the slope of the stable arm and the half-life of the gap. Compare with a Solow economy saving the same rate. Then set $g = 0$ and prove that the speed of convergence falls with σ, with limits as $\sigma \to 0$ and $\sigma \to \infty$.

Solution:

d) (4 points) At $t = 0$ the government unexpectedly and permanently taxes the net return to capital at $\tau = 0.25$, rebating the revenue lump-sum. Find the new $\tilde{k}^*$ and $\hat{c}^*$. Use the linearized stable arm at the new steady state to approximate the jump in $\hat{c}(0)$; the exact value (numerical saddle path) is $\hat{c}(0) = 1.391$. Describe the transition, and say what changes if the tax is announced at $t = 0$ to start at $T > 0$.

Solution:

Block 2 — Money, inflation and seigniorage

(Kurlat chs. 10–11; Romer ch. 11 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) Baumol–Tobin: income $Y = 1200$ a year, cost $F = 2$ per trip to the bank, nominal rate $i = 6\%$. The welfare cost of $i > 0$, measured as the Bailey area under the money-demand curve ($\int_0^i m(x)\,dx - i\,m(i)$), is:

A. 8.49

B. 16.97

C. 4.24

D. 0

b) (3 points) Cagan with perfect foresight, in logs: $m_t - p_t = -a(p_{t+1} - p_t)$, $a = 4$ (monthly, as in Kurlat's Exercise 11.6). At $t = 0$ the central bank credibly announces a permanent 10% rise in the money supply at $t = 4$. On impact, p_0 rises by:

A. 0%

B. 4.1%

C. 10%

D. 8.0%

c) (2 points, T or F) In the Cagan model with perfect foresight, a credible announcement that money growth will fall permanently in the future leaves inflation unchanged until money growth actually falls.

d) (2 points, T or F) If nominal interest income is taxed at 30%, keeping the after-tax real return constant requires the nominal rate to rise by more than one point per point of expected inflation.

Question 4. (15 points) Discursive. Seigniorage and hyperinflation, Brazil in the 1980s (illustrative). Real money demand relative to GDP is $m = \kappa e^{-a\pi^e}$ (κ absorbs the real rate); output is constant; a deficit d (share of GDP) is financed by printing money, $\mu m = d$ with $\mu = \dot{M}/M$; money market clearing gives $\dot{m}/m = \mu - \pi$. Numbers: $\kappa = 0.3$, $a = 2.5$, $d = 0.03$.

a) (3 points) Characterize the steady states. Find the revenue-maximizing inflation and the maximum deficit d_{max}, prove that there are exactly two steady states when $0 < d < d_{max}$, and compute them.

Solution:

b) (4 points) Expectations adapt: $\dot{\pi}^e = \gamma(\pi - \pi^e)$. Derive $\dot{\pi}^e = \frac{\gamma}{1-a\gamma} h(\pi^e)$ with $h(\pi) \equiv \frac{d}{\kappa} e^{a\pi} - \pi$. Prove that if $a\gamma < 1$ the low-inflation steady state is stable and the high one unstable, and that this reverses if $a\gamma > 1$. What happens under perfect foresight? Evaluate with $\gamma = 0.3$.

Solution:

c) (4 points) Let $d > d_{max}$. Prove that $h(\pi) > 0$ for every π, that π^e then rises without bound when $a\gamma < 1$, and that it reaches infinity in finite time. What goes wrong at $a\gamma = 1$?

Solution:

d) (4 points) The deficit rises from 3% to 5% of GDP while the economy sits at π_L, with $\gamma = 0.3$. The time for expected inflation to reach 100% a year is 6.58 years and to diverge, 6.82 years (numerical integration of (c)). Explain where the time goes, using the minimum of h. What does indexation that raises γ above $1/a$ do, and which policies end the process?

Solution: